Protocols – Size-Exclusion Chromatography (FPLC) of DHAK2

TEMPLATE · IGEM 2026 CONTRIBUTION

What you're going to get / What do you expect

*This protocol is a final polishing step for **DHAK2** after IMAC (Protocol_ Immobilised Metal Affinity Chromatography) and dialysis (Protocol_Dialysis_of_IMAC_Eluates). The dialysed sample is run on an **ÄKTAprime plus FPLC system** through a size-exclusion (gel filtration) column. Proteins are separated by size: large aggregates elute first, DHAK2 elutes at its characteristic volume, and smaller contaminants and residual small molecules elute last. The UV absorbance at 280 nm is recorded in real time on the connected computer, so the DHAK2 peak can be seen as it leaves the column.*

The run is isocratic, meaning a single buffer is used throughout, and it takes two days: the system is equilibrated overnight and the sample is injected the next day. The result is a set of 1.5 mL fractions covering the DHAK2 peak, in 50 mM sodium phosphate, 150 mM NaCl, pH 7.2, ready to be tested in the activity assays.

Materials list

Reagents

- Dialysed DHAK2 sample (from the dialysis protocol).
- Sodium phosphate, dibasic: $Na_2HPO_4 \cdot 2H_2O$ (177.99 g/mol).
- Sodium chloride, NaCl (58.44 g/mol).
- HCl and NaOH (1 M), for pH adjustment.
- Ultrapure water (Milli-Q).

Consumables

- 20 mL syringe and needle.
- 5 mL syringe and needle.
- 0.22 µm bottle-top filter.
- 1.5 mL Eppendorf tubes.

Equipment

- ÄKTAprime plus FPLC system, with fraction collector, and a computer with the control software (PrimeView) for real-time monitoring.
- Size-exclusion column: HiLoad 26/600 Superdex 200 prep grade (13mL).
- Syringe adaptor (long connector) for purging the pump.
- Vacuum pump for the bottle-top filter (optional).
- Microcentrifuge.

Steps

A. Buffer preparation:

The running buffer is **50 mM sodium phosphate, 150 mM NaCl, pH 7.2**, and 2L are needed. To avoid weighing and handling such a large volume, prepare a 10× stock and dilute it.

1. Prepare 200 mL of 10× stock: 0.5 M sodium phosphate, 1.5 M NaCl, pH 7.2. Adjust the pH before making up to the final volume.

2. Filter the 10× stock through a 0.22 µm bottle-top filter. Connecting the filter to a vacuum pump makes this much faster. Filtering removes particles that could clog the column and also partly degasses the buffer.
3. Dilute the filtered stock 1/10 with Milli-Q water to a final volume of 2 L.

Note: Keep the 10× stock at room temperature. Concentrated phosphate can crystallise in the cold.

B. System preparation and overnight equilibration (day before the run):

1. Place the inlet tubing in the bottle of running buffer.
2. Purge the system of air every time the mobile phase is connected or removed. We did it the day before the run, because the system was left equilibrating overnight:
    2.1. Unscrew the pump purge connector (*figure 1*) and replace it with the longer connector, which lets you attach the 20 mL syringe.
    2.2. Fill the syringe with Milli-Q water and make sure there are no air bubbles in it. Remove the needle and screw the syringe onto the connector.
    2.3. Inject the water to flush the circuit.
    2.4. Repeat until no air bubbles remain trapped anywhere in the tubing.
    2.5. Remove the syringe and the connector and put the original piece back in place.



*Figure 1. ÄKTAprime plus FPLC system. The **red circle** marks the pump purge connector, which is unscrewed and replaced with a longer connector so that a syringe can be attached to purge air from the system.*

3. Switch on the system. Turn off the UV lamp, because it is not needed during equilibration, and this preserves its lifetime: **Set parameters – OK – Lamp – Off**.
4. Set the equilibration parameters: **Manual run – Set method base (mL) – No gradient – Set flow rate (0.7 mL/min) – Set pressure limit (0.5 MPa)**. The pressure limit protects the column, so check that it does not exceed the maximum pressure stated by the column manufacturer.
5. Check all the parameters, then start the run: **Manual run – Start run – OK – OK**.
6. **Release the pause** by pressing **pause/cont.** on the front panel. The system always starts in pause mode, and the run will never begin unless you release it. ⚠️
7. Leave the system equilibrating overnight. At least 2 column volumes of running buffer must pass through the column. At 0.7 mL/min for ~14 h, about 550–600 mL pass through.

C. System preparation (day of the run):

1. Stop the overnight equilibration: **Stop run**.
2. Check the injection valve in the load position: **Manual run – Set inject valve position – LOAD**.
3. Clean the injection valve circuit:
    3.1. Fill the 5 mL syringe with Milli-Q water and remove any air bubbles. Then remove the needle.
    3.2. Unscrew the black cap of the injection valve (*figure 2*) and screw on the syringe.
    3.3. Inject the water, remove the syringe, and put the cap back.



Figure 2. ÄKTAprime plus FPLC system. The ***blue circle*** *marks the injection valve with its sample loop. The syringe is screwed into the injection port, after its black cap has been removed, to rinse the loop and to load the sample.*

4. Switch on the computer and open the control software to view the chromatogram (UV 280 nm) in real time.
5. Switch on the UV lamp: **Set parameters – OK – Lamp – On**.
6. Set the run parameters: **Manual run – Set method base (mL) – No gradient – Set flow rate (1 mL/min) – Set pressure limit (0.5 MPa)**.
7. Check all the parameters, then start the run: **Manual run – Start run – OK – OK**.
8. **Release the pause** by pressing **pause/cont.** ⚠️
9. Wait until the UV signal is stable, then set the baseline to zero with the autozero function in the Manual run menu (**Manual run – Autozero – OK**). This makes the protein peaks easier to read.
10. *Optional:* stop the run and start a new one after the autozero, so the chromatogram of the sample begins at zero. Before restarting, check all the parameters again and remember to release the pause.

D. Sample injection (day of the run):

1. Clarify the dialysed sample by centrifugation (5 min, max speed) to remove any precipitate that could block the column. Make sure the sample volume does not exceed the volume of the injection loop.
2. Once the UV signal is stable, draw the sample into the 5 mL or 20 mL syringe, depending on its volume. Remove all air bubbles, then remove the needle.
3. Unscrew the black cap of the injection valve and screw the syringe onto it (*Figure 2*).
4. With the valve in the **LOAD** position, push the sample into the loop. Leave the syringe attached, so that the sample cannot flow back out of the loop.
5. Switch the valve to the inject position: **Manual run – Set inject valve position – INJECT**. The running buffer now carries the sample from the loop onto the column. The software marks the moment when the valve changes position, which is the injection point of the chromatogram.
6. After a few minutes, once all the sample has left the loop, switch the valve back to **LOAD** (**Manual run – Set inject valve position – LOAD**). This stops the running buffer from flowing continuously through the injection circuit.

E. Fraction collection (day of the run):

There are two options:

- **Collect fractions from the start of the run.** Use this the first time you run a protein, when you do not know its elution volume yet.
- **Start collecting shortly before the protein elutes.** This uses fewer tubes, but you need to know in advance the elution volume or time of your protein, from a previous run or from the column calibration. We used this option.

1. About 30 min before the expected elution of DHAK2, load the numbered 1.5 mL Eppendorf tubes into the fraction collector.
2. Check that the dispenser arm sits exactly over the first tube and that the rack can rotate freely, so that no sample falls outside the tubes.
3. About 15–20 min before the expected elution, start collecting: **Manual run – Set fraction volume (1.5 mL)**.
4. Follow the chromatogram in real time. Once the DHAK2 peak has fully passed, stop fraction collection.
5. Leave the system running at a lower flow rate (0.6–0.7 mL/min) for a couple of hours, so that any remaining material is washed out of the column.
6. Use the chromatogram to choose the fractions for the activity assays. The fraction numbers on the Eppendorf tubes match those shown in the software. Take all the fractions that make up the peak, so that no protein is lost.

Troubleshooting

- **The run does not start:** The system always starts in pause mode. Press **pause/cont.** after every Start run.

- **Pressure alarm or the run stops on its own:** The pressure limit has been exceeded. Check that the buffer and sample were filtered or centrifuged, that the flow rate is not too high for the column, and that no tubing is kinked or blocked.

- **Air bubbles in the system or a noisy baseline:** Air enters the system whenever the mobile phase is connected or removed. Purge the pump with Milli-Q water every time you change the mobile phase, and also flush the injection valve, until no bubbles remain in the tubing. Use filtered, degassed buffer, and remove all air from the syringes before connecting them.

- **Drifting baseline:** Wait until the UV signal is stable before doing the autozero and injecting the sample.

- **Sample lost before reaching the column:** The sample may have flowed out of the loop. Keep the syringe attached after loading, and do not load more volume than the loop can hold.

- **DHAK2 peak missed or not where expected:** The first time, collect fractions from the start of the run to locate the peak, and note its elution volume for later runs.

- **Very small or no protein peak:** The sample may be too dilute or protein may have been lost during dialysis. Measure the protein concentration before injecting it.